

Harsh Soni

+1 (352) 328-4717 | harshsoni.06@outlook.com | linkedin.com/in/harshh06 | github.com/harshh06 | [Portfolio](#)

Summary

Full-stack and AI engineer with **3 years** at Attentive.ai and Infosys. Shipped scalable systems serving **500+ enterprise clients**. MS CS at UF; strong in **Python, React, FastAPI**, and production **LLM/RAG pipelines**.

Education

University of Florida, Gainesville, FL

Aug 2025 – May 2027

Master of Science in Computer Science · GPA: 3.8

Relevant Coursework: Distributed Systems, Machine Learning, Advanced Data Structures, Operating Systems

Jaypee University of Engineering and Technology, India

Jul 2018 – Jun 2022

Bachelor of Technology in Computer Science and Engineering · GPA: 3.4/4.0 · Top 15%

Work Experience

Attentive.ai, Noida, India

Jun 2023 – Jun 2025

Software Development Engineer | Skills: TypeScript, React, Python, PostgreSQL, Redux, Next.js

- Diagnosed a critical 12s settings-page load bottleneck impacting **500+ enterprise clients**; independently owned and shipped a layered fix using React code-splitting, debounced zoning calculations, and IndexedDB caching, reducing load time by **86%** (to 1.7s) and directly reducing client churn risk.
- Identified slow construction document Q&A as a top client pain point and **independently built Beam GPT** end-to-end: an LLM-powered assistant using **RAG** (Pinecone vector DB + OpenAI GPT-4) that cut task completion time by **60%** for **500+ users**, with chunked ingestion and semantic reranking for low-latency retrieval.
- Built a hierarchical folder system on the frontend enabling **1,000+** takeoff documents per client, implementing optimistic UI updates that reduced folder navigation latency by **40%**.
- Consolidated **8 fragmented data sources** by owning a full rebuild of the CSM Dashboard and **RESTful API** integration, boosting team productivity by **30%**.
- Redesigned Feature Panel with **TanStack Virtual** windowing and **Intersection Observer** for viewport-based rendering, lazy-loading via **React.lazy** + **Suspense** and **Next.js dynamic imports**, handling **GB-scale** data in the browser.

Infosys, Bangalore, India

Feb 2022 – Jun 2023

Software Engineer (Specialist Programmer) | Skills: Python, Flask, React, Java, PostgreSQL, Redis

- Migrated a CLI-based internal tool to a React/Flask web app for **Apple IS&T**, replacing manual workflows with a browser UI backed by **Redis** caching to reduce repeated query latency.
- Contributed **5 merged PRs** to **neo4j-graphql-java** resolving **race conditions** in the query executor and hardening GraphQL schema validation, improving concurrency correctness across **3 internal teams**.

Projects

Automated Multi-Agent PR Reviewer (PR Sentinel)

[GitHub](#)

FastAPI, LangGraph, Celery, Redis, Pinecone, Google Gemini, PyGithub, GCP Cloud Run

- Deployed a production **GitHub App** on **GCP Cloud Run** authenticating via JWT tokens and verifying **HMAC-SHA256** webhook signatures, posting consolidated AI reviews on PR threads with zero human intervention.
- Architected a **multi-agent RAG pipeline** using **LangGraph** spawning **3 parallel async sub-agents** (Security, Performance, Code Quality) with autonomous multi-turn tool-calling loops over a **Pinecone** vector index; orchestrator aggregates findings into a structured Markdown review.
- Engineered a durable **two-stage Celery/Redis pipeline** (ingest → review) with exponential backoff retries (10× over ~30 min), rate-limit parsing, atomic quota enforcement, and SHA-scoped idempotency keys to prevent duplicate webhook processing.
- Designed **incremental vector ingestion** using per-repo **Pinecone namespaces**, re-embedding only changed files per PR and deleting removed file vectors, reducing ingestion time by **~90%**.

AI-Powered Histopathology Scoring System

[GitHub](#) | [Live Demo](#)

Next.js, FastAPI, PyTorch, ResNet18, PostgreSQL, Docker, Hugging Face, GitHub Actions

- Built a diagnostic tool for pancreatic tissue analysis by fine-tuning **ResNet18** (transfer learning on ImageNet weights), achieving **4-metric** pathology scoring and reducing manual analysis time by **40%**.
- Engineered a client-side upload queue with a **FIFO buffer** to handle **50MB+ TIFF batches**, preventing server RAM exhaustion by throttling concurrent FastAPI requests to 3 via **async/await**.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript (ES6+), C++, Java, SQL, HTML/CSS
- **Frameworks & Libraries:** React, Next.js, Redux, FastAPI, Flask, Node.js, PyTorch, LangGraph, LangChain, Tailwind CSS
- **Tools & Platforms:** Docker, Git, PostgreSQL, MongoDB, Redis, Pinecone, Cloud (AWS Amplify/CloudFront, GCP Cloud Run), GitHub Actions, Linux
- **Concepts:** REST API Design, Microservices, CI/CD, Agile/Scrum, Scalability, Low-Latency System Design, LLMs, RAG, Vector Databases, Embeddings, Machine Learning, OOP